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IARCO RESEARCH PROPOSAL

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Research topic: From Inhaled Tobacco and Nicotine Exposure to Vascular Damage: A Systematic Review of Their Effects on Endothelial Function, Arterial Stiffness, and Recovery After Quitting.

Title: From Inhaled Tobacco and Nicotine Exposure to Vascular Damage: A Systematic Review of Their Effects on Endothelial Function, Arterial Stiffness, and Recovery After Quitting.

Introduction and Research Problem

Tobacco and nicotine exposure are the major risk factors that contribute to cardiovascular disease.The inner lining of blood vessels called endothelium is a very thin layer of cells that plays an important role in maintaining blood flow and vascular function, consequently impaired endothelial function provides early indication of vascular injury. Previous studies have associated tobacco smoking with oxidative stress, inflammation, reduced (NO) levels and reduced flow mediated dilation(FMD) all linking to endothelial dysfunction. Hahad et al. identified increased oxidative stress and reduced NO levels due to tobacco exposure as an important factor that contributes to vascular damage.

Key explanation for this damage is the interaction of reactive oxygen species (ROS) and (NO). tobacco exposure can increase oxidative stress leading to excess ROS and reduced NO availability in the blood vessel that impairs vasodilation (widening of the vessel). Inflammation also contributes to endothelial damage. These effects occur with different inhaled nicotine and tobacco products, not only cigarettes. Belkin et al. investigated cigarettes, heated tobacco products and e cigarettes in relation to inflammation and endothelial dysfunction.[1] Likewise, Rezk- Hanna et al found that even a short exposure of nicotine containing e hookah in healthy adults reduces flow mediated dilation (FMD) and causes oxidative stress by increasing ROS level and decreasing NO levels. [6]

Vascular damage can also be measured by arterial stiffness, that is measured using pulse wave velocity (PWV), higher PWV indicates greater arterial stiffness. A systematic review from previous studies found that smoking and vaping causes an increase in PWV, while smoking cessation reduces arterial stiffness. [2] Evidence from previous studies also supports possible biological mechanisms. Zaaba et al. found that waterpipe smoke caused oxidative stress, inflammation and cellular damage in human endothelial cells. [3] However this was an in vitro study and these findings provide mechanistic rather than direct effect.

However, an important unresolved issue is vascular recovery after cessation. If tobacco and nicotine exposure contributes to vascular damage, it is important to understand that these changes can improve after quitting. Existing studies do not show a uniform pattern. For example, one of the previous studies by Okuyama et al found improvement in flow mediated total dilation (FMTD) after 20 weeks of quitting smoking, while the difference in FMD itself was not significant. [8]

Thus the problem is not simply if tobacco and nicotine exposure damages blood vessels. The existing research is spread across different products (cigarettes , e-cigarettes, vaping, e-hookah ,tobacco) the biological mechanisms and vascular measurements, while evidence on

the extent and pattern of recovery after cessation remains less consistent. So, an integrated review is needed to connect the pathway from exposure and biological changes to vascular damage and recovery after quitting.

Existing Literature and Research Gap

Current Research shows how tobacco and nicotine exposure causes vascular damage through oxidative stress, inflammation and reduced (NO) levels. Hahad et al. showed human evidence explaining this pathway, Kumboyono et al. concluded decrease in NO levels and increase in malondialdehyde (MDA) levels in smokers as compared to non smoker. [5], [4] However it doesn't shows changes overtime or recovery after quitting. Current studies also used vascular measures to support these effects. Nicotine containing e-hookah reduced FMD levels with changes in ROS and NO levels. [6] A systematic review and meta analysis also found that PWV levels in smokers and vapes were higher and they decreased after cessation. [2] However, results for vascular recovery after quitting are not consistent. Kolnizakis et al. discovered improved FMD in the first 6 months after quitting but participants underwent additional interventions during those 6 months of quitting. [7] Okuyama et al. found improved FMDT after 20 weeks of quitting but change in FMD levels was not significant. [8] Thus, the research gap is the lack of research evidence comparing different inhaled products and their vascular outcomes and connecting it to how vascular function changes after cessation.

Research Question

How do inhaled tobacco and nicotine containing products affect endothelial function and arterial stiffness through oxidative stress, inflammation and reduced NO availability, and what does current evidence indicate about vascular recovery after quitting?

objectives:

the review will:

1. Investigate how inhaling tobacco or nicotine affects endothelial function, by looking at changes in FMD levels.
2. It will examine how tobacco and nicotine usage causes increase in arterial stiffness by measuring PWV levels
3. Explore the biological mechanisms linking tobacco and nicotine exposure to vascular dysfunction by examining evidence on oxidative stress, inflammation and NO availability.
4. Examine how vascular health changes after a person stops using these products, and if the damage is reversible.
5. Compare these effects according to product type(cigarettes, vapes, hookah etc), levels of exposure and outcomes and duration after quitting.

Hypothesis:

based on existing literature this review expects to find that inhaling tobacco and nicotine will increase oxidative stress and inflammation and will reduce NO levels. This may contribute to endothelial dysfunction and arterial stiffness. Quitting is hypothesized to result in partial vascular improvement still the speed and extent of recovery will depend on the type of product and its intensity and the individual characteristics.

Methodology

1. Study design:

This will be a systematic review focusing on human studies with relevant in-vitro studies included separately as mechanistic evidence that will follow the PRISMA 2020 reporting guidelines.

2. Literature Search:

PubMed will be searched using 4 key concepts:

- Exposure or type of product: smoking, tobacco, cigarette, e cigarette, vaping, heated tobacco, waterpipe smoking, hookah
- Effects that will be used for measurements: endothelial dysfunction and endothelial function, flow mediated dilation, arterial stiffness, Pulse wave velocity.
- Mechanism: oxidative stress(ROS), nitric oxide(NO), inflammation.
- Recovery part: cessation, quitting, recovery.

These steps will be combined using boolean operators and the final search key words search date and number of results will be recorded in a document so that it can be reproduced.

3. Eligibility criteria:

The review will include results on human adults exposed to cigarettes, e cigarettes, vapes, heated tobacco, water pipe or e hookah. Studies will be included if they are on endothelial function, arterial stiffness, inflammation or levels of NO or recovery after quitting. Studies will be excluded that will be animal related, unrelated exposure, case reports, duplicate publications and studies without relevant outcomes. Cannabis or edible products studies will not be included and relevant in vitro studies will be kept only for mechanistic reference.

4. Study Selection and Data Extraction

After removing duplicate titles and abstracts would be checked for the eligibility criteria then check full text of potential eligible studies and reasons would be recorded for excluding full text studies. The selection process will be shown using the prisma 2020 flow chart diagram after the review is completed.

Data will be extracted on author, year, study design, population, product used, exposure duration, vascular outcome, measure of FMD NO ROS PWV will be recorded. FMD and

PWV will be the main focus as these levels are the main measures for endothelial dysfunction and arterial stiffness.

5. Risk of Bias and Data Synthesis

Risk of bias will be assessed using tools appropriate for each study design. Information will then be grouped by product type, biological mechanism, vascular outcome and quitting duration. A quantitative meta-analysis will only be done if the results are uniform in population, exposure and effect and outcomes. if there will be too much clinical or methodological difference in findings it will be described narratively instead of meta analysis. This allows the review to clearly show the difference between immediate effects and long-term vascular damage, and recovery after quitting rather than treating all outcomes as equal.

Expected Significance and Innovation

This review connects biological processes of oxidative stress, inflammation, and reduced NO levels to real-world blood vessel measurements like FMD and PWV. It also keeps human study results separate from lab experiments. This ensures we understand the biology without exaggerating how it applies to actual patients. Comparing different inhaled tobacco and nicotine products can show whether they cause similar or different types of blood vessel damage. Additionally, by looking at both FMD and PWV levels we can see if different parts of the vascular system heal at different speeds. Current evidence of previously done studies already suggests that recovery is not uniform across all vascular measures. [2], [8].

In summary, this review provides a comprehensive look at blood vessel damage and recovery from inhaled tobacco and nicotine exposure. It connects the pathway from exposure to endothelial dysfunction and arterial stiffness and also examines evidence for recovery after quitting.

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